

STATE OF THE ART · THESIS MORO

Ultrasound Simulators for Robot-Assisted Breast Biopsy

Systematic review of B-mode ultrasound simulation methods for generating images from CT volumes — classified by simulator family, assessed against the thesis hardware and real-time constraints.

PAPERS

32 curated

METHOD

Scopus systematic search

FAMILIES

Wave · Ray · Convolution · Learning

THEMATIC BLOCKS

6 + feasibility

Università di Trento · DISI · Supervisors: E. Lamon (robotics) · S. Tomassini (medical imaging) · Reference taxonomy: Solano-Cordero et al., MDPI Applied Sciences 2025

01

Foundational simulators (pre-2014)

SEMINAL REFERENCES

Papers cited by every modern simulator — define the convolution / wave-based / ray-based families.

SCOPUS TITLE-ABS-KEY (("ultrasound simulation" OR "simulated ultrasound" OR "B-mode simulation" OR "k-Wave")) AND PUBYEAR < 2014

#	Authors, Year	Title	Source	Family	Summary	DOI
1.1	Bamber & Dickinson 1980	Ultrasonic B-scanning: a computer simulation	Phys. Med. Biol. 25(3)	Convolution	First numerical simulation of a clinical B-mode image. Tissue as point scatterers, echo via 1-D PSF convolution. Defines the convolution family all CT-to-US methods build on.	10.1088/0031-9155/25/3/004
1.2	Jensen 1996	Field: A program for simulating ultrasound systems	Med. & Biol. Eng. & Comp. 34	Wave	Foundational Matlab simulator using spatial impulse responses for arbitrary transducer geometries. De-facto reference cited by virtually every later simulator paper.	–
1.3	Jensen 2004	Simulation of advanced ultrasound systems using Field II	IEEE ISBI 2004	Wave	Field II update for synthetic-aperture and coded excitation. 32-CPU cluster cuts sim time from 17 months to 391 h. Modern citation key for Field II.	10.1109/ISBI.2004.1398618
1.4	Shams, Hartley, Navab 2008	Real-Time Simulation of Medical Ultrasound from CT Images	MICCAI 2008 LNCS 5242	Convolution	Splits CT-to-US into one-shot pre-processing (3D scattering volume from segmented CT) + run-time plane tracing with acoustic artefacts. Canonical CT → B-mode reference.	10.1007/978-3-540-85990-1_88
1.5	Wein et al. 2008	Simulation and fully automatic multimodal registration of medical ultrasound	Med. Image Analysis 12(5)	Convolution	Derives an acoustic-impedance proxy from CT Hounsfield Units, convolves with a depth-dependent PSF to produce B-mode that registers to real US. Underpins the CT-to-US literature with Shams 2008.	10.1016/j.media.2008.06.006
1.6	Treeby & Cox 2010	k-Wave: MATLAB toolbox for simulation and reconstruction of photoacoustic wave fields	J. Biomedical Optics 15(2)	Wave	Open-source time-domain full nonlinear wave propagation in heterogeneous media. k-space pseudospectral, CPU/GPU. Standard wave-based reference — the engine behind the MUSiK evaluation.	10.1117/1.3360308
1.7	Bürger et al. 2013	Real-Time GPU-Based Ultrasound Simulation Using Deformable Mesh Models	IEEE Trans. Med. Imaging 32(3)	Ray	GPU ray-tracing over deformable meshes (>2M triangles) with PhysX deformation. Models attenuation, refraction, shadowing in real time. The architecture NVIDIA Isaac modernises with OptiX in 2025.	10.1109/TMI.2012.2234474

02

Ray-based & Scatterer-based (post-2014)

BASE FOR NVIDIA ISAAC

Compromise between physical accuracy and real-time performance.

SCOPUS TITLE-ABS-KEY (("ray tracing" OR "raytracing" OR "Monte Carlo") AND ultrasound AND (simulation OR synthesis)) AND PUBYEAR > 2015

#	Authors, Year	Title	Source	Family	Summary	DOI
2.1	Mattausch O. 2017	Comparison of texture synthesis methods for content generation in US simulation for training	SPIE Med. Imaging Proc.	Ray/texture	Compares three procedural texture-synthesis strategies for micro-structure (speckle, fibrosis, fat) inside a ray-traced simulator. Reference for the texture-on-organ step any ray pipeline needs.	10.1117/12.2250604
2.2	Mattausch O. 2018	Realistic US simulation of complex surface models using interactive Monte-Carlo path tracing	Computer Graphics Forum	Ray	Interactive Monte-Carlo path tracing on triangle-soup meshes. Adds multiple scattering, reverberation, refraction that Bürger 2013 could not. Immediate predecessor of the 2025 Isaac generation.	10.1111/cgf.13260
2.3	Satheesh B.A. 2018	A method of US simulation from patient-specific CT image data: preliminary study	IEEE ISBI 2018	Scatterer/CT	Each CT voxel → scatterer population with density/amplitude from Hounsfield value, then Field-II-like beamforming. Direct ancestor of the workflow envisaged for this thesis.	10.1109/ISBI.2018.8363853
2.4	Starkov R. 2019	Ultrasound simulation with deformable and patient-specific scatterer maps	IJCARS	Scatterer	Builds patient-specific scatterer maps from MR/CT, deforms them with FEM, then interrogates them to render US. Bridges surface ray simulators and volumetric scatterer simulators à la Field II.	10.1007/s11548-019-02054-5
2.5	Satheesh B.A. 2019	A fast method for simulating US image from patient-specific CT data	Biomed. Signal Proc. Control	Scatterer/CT	GPU-accelerated convolution kernel cuts per-frame sim from minutes to seconds. Validated on liver phantoms + Visible Human CT. Closest published method to the breast-phantom goal.	10.1016/j.bspc.2018.10.003

State of the art post-2018 — learning appearance from real data instead of physics.

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SCOPUS TITLE-ABS-KEY ( "ultrasound simulation" OR "ultrasound image synthesis" OR "synthetic ultrasound" OR "B-mode simulation" ) AND PUBYEAR > 2014
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#	Authors, Year	Title	Source	Family	Summary	DOI
3.1	Hu Y. 2017	Freehand US image simulation with spatially-conditioned GANs	LNCS / MICCAI	GAN	UCL condition a GAN on 6-DoF probe pose for spatially consistent freehand sweeps. Early work on temporal coherence — essential for any robotic application streaming images.	10.1007/978-3-319-67564-0_11
3.2	Tom F. 2018	Simulating patho-realistic US images using deep generative networks	IEEE ISBI 2018	GAN seminal	First GAN-based US simulator injecting realistic pathology (cysts, lesions) into healthy images. Introduces the patho-realistic GAN concept later extended by spGAN, ConPres.	10.1109/ISBI.2018.8363780
3.3	Peng B. 2019	A Real-Time Medical US Simulator Based on a GAN Model	IEEE ICIP 2019	GAN real-time	First real-time (~15 fps) GAN US simulator on a single GPU. Targets sonographer training, but architecture reusable for a fast in-the-loop synthesiser.	10.1109/ICIP.2019.8803570
3.4	Bargsten L. 2020	SpeckleGAN: GAN with an adaptive speckle layer to augment limited training data	IJCARS	GAN speckle	Adds a physics-inspired speckle layer so the network learns appearance and speckle separately. Augmented images improve downstream segmentation when data is scarce.	10.1007/s11548-020-02203-1
3.5	Gilbert A. 2021	Generating Synthetic Labeled Data from Existing Anatomical Models (echo segmentation)	IEEE Trans. Med. Imaging	GAN	KCL generate paired image+segmentation data: simulate echo from anatomical models then translate with CycleGAN. Closes the physics ↔ learning loop — a template the thesis can reuse.	10.1109/TMI.2021.3051806
3.6	Liang J. 2022	Sketch guided and progressive growing GAN for editable US image synthesis	Medical Image Analysis	GAN sketch	Most-cited modern US-synthesis paper (123 cit). Editable anatomical sketch → realistic US following it, with iterative lesion/organ shape editing. Relevant for dataset augmentation.	10.1016/j.media.2022.102461
3.7	Stojanovski D. 2023	Echo from Noise: Synthetic US Generation Using Diffusion Models	LNCS	Diffusion	First diffusion-model US generator, conditioned on segmentation masks. Beats GAN baselines on downstream segmentation; inaugurates the diffusion family (2023+).	10.1007/978-3-031-44521-7_4
3.8	Wysocki M. 2023	Ultra-NeRF: Neural Radiance Fields for Ultrasound Imaging	PMLR	NeRF	TUM adapt NeRF to US, replacing the optical rendering equation with an acoustic one. Sparse posed sweeps → continuous volumetric representation, enabling sim-to-real.	–
3.9	Freiche B. 2025	Ultrasound Image Generation using Latent Diffusion Models	SPIE Med. Imaging Proc.	Diffusion	Latent diffusion for conditional/unconditional US generation. Higher fidelity than pixel-space diffusion at lower compute. One of the most recent works (2025).	10.1117/12.3047788

04CT-to-US translationPRIMARY THESIS WORKFLOW

From a patient/phantom CT volume to a simulated B-mode image — the thesis input/output structure.

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SCOPUS TITLE-ABS-KEY ( ( "ultrasound simulation" OR "simulated ultrasound" ) AND ( "CT" OR "computed tomography" OR "CT-derived" OR "CT-based" ) ) AND PUBYEAR > 2008
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#	Authors, Year	Title	Source	Family	Summary	DOI
4.1	Song Y. 2024	<i>S-CycleGAN: Semantic Segmentation Enhanced CT-US Translation for Robotic Ultrasonography</i>	IEEE Conf. 2024	CT-US robotic	Augments CycleGAN with a semantic-segmentation loss so structures are preserved during CT-to-US translation. Targets robotic scanning — most operationally similar published pipeline to the thesis.	10.1109/CBS61689.2024.10860598
4.2	Wang X. 2025	<i>CT to US abdominal image synthesis using domain-constraint GANs</i>	Biomed. Signal Proc. Control	CT-US	Domain-constrained GAN preserving anatomy across modalities. Found in 3/4 Scopus queries — the most cross-cutting recent CT-to-US paper, directly aligned with the thesis structure.	10.1016/j.bspc.2025.108344
4.3	Zhang C. 2025	<i>CT-based radial endobronchial US image simulation of peripheral pulmonary lesions using deep learning</i>	Endoscopic Ultrasound	CT-US	Generates radial-EBUS images from preoperative CT for procedure planning. Shows clinical viability of CT-to-US simulation in a niche modality; validated against real intra-op scans.	10.1097/eus.000000000000079

05Breast-specific simulatorsTHE BREAST NICHE

Few works exist — virtual-breast-project is the recognised reference (linked by the supervisor).

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SCOPUS TITLE-ABS-KEY ( ultrasound AND ( simulation OR synthesis ) AND ( breast OR mammary ) AND ( phantom OR model OR virtual ) ) AND PUBYEAR > 2010
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#	Authors, Year	Title	Source	Family	Summary	DOI
5.1	Wang Y. 2015 ★★	<i>Building a virtual simulation platform for quasistatic breast US elastography (open source)</i>	Medical Physics	Breast sim	Michigan Tech chain Field II + Tetgen + FEBio + VTK to build a virtual breast (skin, fibro-glandular, Cooper ligaments, ducts, tumour), deform under FE compression, simulate B-mode RF. Gold-standard breast simulator for offline validation.	10.1118/1.4928707
5.2	Jiménez-Gaona Y. 2024	<i>GAN-based data augmentation to improve breast US and mammography mass classification</i>	Biomed. Signal Proc. Control	Breast GAN	GAN-augmented breast US datasets improve mass classification by 6–8% AUC. Evidence that synthetic breast US images carry clinical training value — supports the thesis claim.	10.1016/j.bspc.2024.106255
5.3	Zama S. 2024	<i>Clinical Utility of Breast US Images Synthesized by a GAN</i>	Medicina (Lithuania)	Breast GAN	Blind expert review of GAN-synthesised breast US: 84% rated clinically realistic. Provides a published acceptability bar to later measure the thesis simulator against.	10.3390/medicina60010014

BRIDGE TO APPLICATION

#	Authors, Year	Title	Source	Family	Summary	DOI
6.1	Zhu M. 2011	Real-time image-based B-mode US simulation of needles using tensor-product interpolation	IEEE Trans. Med. Imaging	Robotic/needle	UBC simulate B-mode of inserted needles via tensor-product interpolation of pre-recorded templates. Critical for the biopsy phase where the simulator must depict a needle in real time.	10.1109/TMI.2011.2121091
6.2	Abadi E. 2020	Virtual clinical trials in medical imaging: A review	J. Medical Imaging	Review	Most-cited (215) review of virtual clinical trials with a substantial US section. Frames why high-fidelity simulators matter for evaluating imaging techniques without real patients.	10.1117/1.JMI.7.4.042805
6.3	Solano-Cordero C.M. 2025 ★	Recent Advances in B-Mode Ultrasound Simulators	Applied Sciences (MDPI)	★ Review	Scoping review of 80 studies (2014–2024), structured search across PubMed/Scopus/WoS/IEEE. Organises simulators into wave / ray / convolution / AI families — the classification adopted by this thesis.	10.3390/app152312535
6.4	Li Y. 2025	UltrAP-Net: Reverse Approximation of Tissue Properties in US Imaging	Advanced Intelligent Systems	Robotic/props	TUM invert the imaging equation: real US → underlying acoustic tissue properties, which then drive a forward simulator. Bridges learning and physics; same group as Ultra-NeRF.	10.1002/aisy.202400865
6.5	Chan T.J. 2026 ★★	MUSiK: An Open Source Simulation Library for 3D Multi-View Ultrasound	IEEE Trans. Biomed. Eng.	Open 3D	Penn release a fully open-source library for 3-D multi-view US from any anatomical volume. Found in 4/4 Scopus queries (highest cross-cutting). Evaluated as a primary candidate — see feasibility table.	10.1109/TBME.2025.3605577

RTX 2000 ADA · 8 GB · REAL-TIME

● Real-time ● Conditional ● Not real-time ▶ Primary candidate ▶ Viable ▶ Blocked / discouraged

Decision narrative for the defense. The thesis followed a principled funnel: (1) **MUSiK / k-Wave** was the highest-scoring open candidate and was fully implemented and benchmarked, but its wave-solver imposes a hard physical floor of ~220 s per acquisition — structurally incompatible with the real-time control loop. (2) **NVIDIA Isaac standalone (raysim, OptiX)** was adopted: it isolates the GPU raytracer from the unusable full workflow and runs on the 8 GB laptop at ~5 ms/frame. The aim, confirmed by the supervisor, is **lesion localisation**, not diagnostic-grade imaging — which the ray-tracing fidelity comfortably supports. Field II and virtual-breast-project remain as offline physical ground truth.

